

Ransomware with Internet of Things (IoT)

Carson Blin

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1 Reviewed Literature

For my project topic, I will be looking into ransomware attacks on IoT devices. Throughout this second literature review process, I have seen what other researchers have done in regards to ransomware and the IoT field. The findings specifically regarding ransomware and IoT are below:

- Ibrar Yaqoob et al., “The rise of ransomware and emerging security challenges in the Internet of Things,” Special Issue on 5G Wireless Networks for IoT and Body Sensors, *Computer Networks* 129 (2017): 444–458, ISSN: 1389-1286, <https://doi.org/https://doi.org/10.1016/j.comnet.2017.09.003>, <https://www.sciencedirect.com/science/article/pii/S1389128617303468>
- Mamoon Humayun et al., “Internet of things and ransomware: Evolution, mitigation and prevention,” *Egyptian Informatics Journal* 22, no. 1 (2021): 105–117, ISSN: 1110-8665, <https://doi.org/https://doi.org/10.1016/j.eij.2020.05.003>, <https://www.sciencedirect.com/science/article/pii/S1110866520301304>
- Muna Al-Hawawreh, Frank den Hartog, and Elena Sitnikova, “Targeted Ransomware: A New Cyber Threat to Edge System of Brownfield Industrial Internet of Things,” *IEEE Internet of Things Journal* 6, no. 4 (2019): 7137–7151, <https://doi.org/10.1109/JIOT.2019.2914390>
- Amjad Alraizza and Abdulmohsen Algarni, “Ransomware Detection Using Machine Learning: A Survey,” *Big Data and Cognitive Computing* 7, no. 3 (2023), ISSN: 2504-2289, <https://doi.org/10.3390/bdcc7030143>, <https://www.mdpi.com/2504-2289/7/3/143>

2 Summary of Learning

Throughout these research papers, I learned more information specifically about ransomware and the social engineering tactic used to persuade users into paying the ransom. An example of this is scareware, which tactically scares the user

using flashy colors, loud noises, and popups to persuade the user into downloading malicious software. This software can be commonly masked as an antivirus software, but in reality it is a malicious application and can be very dangerous. I also learned how ransomware is broken up into different levels of ransomware. The levels go as follows: Ransomware is distributed across emails with attachments, the user downloads the attachment and gets infected, the malicious payload is then in process, scans the user's device, encrypts the data making it unavailable, and then the user will be prompted to pay to decrypt their data. This information is important to the overall scope of the project solely due to how ransomware works and how it can infect users as simple as one click of a button. As IoT devices get larger in terms of scale, the overall security of IoT has to increase and be more strict.

3 Selected Security Problem

The security problem I want to address is regarding the distribution of ransomware and how to prevent ransomware from gaining further access into the device or business. This is important because it is vital to find the ransomware early and quarantine the source as soon as possible. One of the reasons that ransomware is able to infect many user's device and businesses so easily is due to how easy it is for an individual to mistake a legitimate email from a malicious one. From a security perspective, this is a serious matter and can have catastrophic financial impact for the user and company.

4 Next-Phase Plan

In my next report, my next few objectives are to strategically plan out effective methods and overall approach for this project. I also want to see what kind of datasets are available regarding ransomware attacks to help gauge a sense of scale. I also want to perform an analysis and find a trend in what ransomware has been the most popular within the last 5 years and based on what kind of devices were infected.